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Trajectory planning for AGV based on the improved artificial potential field- A* algorithm

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Abstract

There are many redundant nodes and inflection points in the path planned by the traditional A* algorithm, leading to the inefficient trajectory planning of the automatic guided vehicle (AGV) in the multi-static obstacles environment. The artificial potential field (APF) algorithm suffers from the problem of unreachable objectives and falling into optimal local value. This article studies the trajectory optimization of AGVs to improve the trajectory planning algorithm's iteration efficiency and shorten the trajectory's total length. This article establishes the forward kinematic and unified robot description format model of the AGV and proposes the APF-A* algorithm for trajectory planning. The search cost and the number of turns are effectively optimized. The article simulates the APF-A* algorithm, the results are compared with the trajectory before optimization, and the optimized time is 60% less than that before optimization. The experimental platform of AGV trajectory planning is built, and the algorithm verification experiment of AGV trajectory planning is carried out. The experimental results show that the algorithm studied in this article achieves path smoothing and trajectory length optimization.

Keywords: automatic guided vehicle, trajectory planning, APF-A* algorithm, redundant nodes, unified robot description format model

1. Introduction

The intelligent storage system is a new automated system established based on artificial intelligence technology. The most typical feature is introducing a large number of automatic guided vehicles (AGVs) [1, 2] in warehouses to assist in cargo selection and handling goods. Therefore, AGV has become one of the research objects of famous scholars at home and abroad. The trajectory planning of AGV guarantees safe and accurate operation, and obtaining accurate and real-time trajectory information is an essential factor and fundamental prerequisite to realizing high-efficiency operation. Planning the optimal trajectory accurately, quickly, and safely is a critical problem that needs further research. Many domestic and foreign scholars have conducted extensive research on this.

There are two approaches to the study of global trajectory planning for AGV. One type is localization mapping [3, 4], where the vehicle is placed in a new environment by applying simultaneous localization and mapping (SLAM) [5]. During vehicle traveling, sensors such as cameras or LIDAR [6, 7] automatically scan the relative positions of various elements in the surrounding environment, such as obstacles and the corresponding 3D structure of the ground surface. In this way, a map of the intelligent factory is created incrementally. In response to the requirement of high localization accuracy of AGV in industrial scenes. Li *et al* [8] proposed an improved visual SLAM (VSLAM) algorithm. Lin *et al* [9] used Kalman filtering to remove Gaussian noise to improve the global positioning accuracy of AGV. The above

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two methods cannot accurately plot the travel route and establish an estimation-type model, which is challenging to meet the demand of AGV global trajectory planning for AGV to draw smooth paths and avoid falling into local optimum traps. Zhang *et al* [10] proposed a novel hybrid algorithm with improved ant colony optimization. Abdurrahman Yilmazetet and Temeltas [11] improved self adaptive Monte Carlo localization for particle filter coarse localization.

The other one is scanned and identified by multiple sensors and passed to the upper machine for global trajectory planning through algorithms [12, 13]. Currently, many scholars have proposed methods applied to vehicle trajectory planning, such as fuzzy logic algorithm [14, 15], tabu search algorithm [16, 17], and artificial potential field (APF) method [18]. However, with most of the above planning methods, developing efficient and stable planning rules is challenging, and poor strain is bound to cause problems such as inefficient trajectory planning and unreachable goals. For the problems of AGV trajectory planning efficiency and target unreachable. Wang *et al* [19] used the improved A* algorithm to solve the problems of low trajectory planning efficiency, large memory resource consumption, and low trajectory smoothness. Guan *et al* [20] proposed a Chebyshev distance weighting method for the heuristic function of the traditional A* algorithm. This method significantly reduces the search time and the number of search nodes of the A* algorithm and improves the path search efficiency of the traditional A* algorithm. Yulan and Nannan [21] proposed a base algorithm of Dijkstra for AGV path optimization based on the ant colony algorithm and introduced a time interval factor for optimization. However, all of them only consider the problems of local minimum and target unreachability and have yet to fully consider the problems of removing redundant nodes and extra inflection points of trajectories. Li *et al* [22] proposed a robot trajectory planning method based on improved A* and dynamic window approach (DWA) fusion algorithms (DWA-A*). Based on this, the obstacle avoidance algorithm is implemented. However, the iterative function is significantly more computationally intensive. Luo *et al* [23] proposed an AGV global trajectory planning solution model based on the deep Q-networks (DQN) algorithm. However, the solution function model of DQN is sparse, and the search space is enormous. The method needs to improve its search efficiency, slow convergence speed, and even failure to converge. To address the problems of low positioning accuracy and significant trajectory deviation when AGV works in dynamic environments. Yuan *et al* [24] proposed an unscented Kalman filter (UKF) and adaptive Monte Carlo localization algorithm for an ultra-comprehensive band positioning system. Xiaoming *et al* [25] proposed an adaptive UKF-based optimal positional posture estimation method for following AGV.

However, many perceptual environment and trajectory planning studies ignore the local optimal solution problem and have yet to fully consider the trajectory quality, smoothing, and other issues. At the same time, the DWA-A* algorithm needs to analyze the target point azimuth in real time to distinguish obstacles. The DWA-A* algorithm has problems such as redundant inflection points and many iterations during operation. In this article, an improved APF-A* algorithm is proposed to optimize the heuristic function by taking AGV as the research object. The distance compromise value is selected as the estimation value of the heuristic function. Combining the advantages of the APF, we first consider increasing the offset cost of pre-selected nodes to the target nodes. Then, when two adjacent nodes have exact costs, adjust the repulsion of obstacles to the turning nodes and optimize the iterative strategy to reduce the computation. Achieve the shortest time to find the minimal cost trajectory on the road network map. The main contributions are as follows:

This paper proposes a hybrid algorithm with an improved A* algorithm and APF method.

Firstly, the repulsive field function of the APF method is improved to avoid falling into the local minimum and improve the security of the planning trajectory.

Then, based on the traditional A* algorithm, the potential field is incorporated into the dynamic weight coefficients of the improved A* algorithm, and the dynamic weight coefficients are added to the cost function to improve the algorithm's search efficiency.

The rest of this article is organized as follows. Chapter 2 establishes the AGV forward kinematic model. Chapter 3 fuses the A* algorithm with the APF method. Chapter 4 uses the APF-A* algorithm for AGV trajectory planning simulation analysis-chapter 5 experiments. Finally, conclusions are drawn in Chapter 6.

2. AGV kinematic model establishing and unified robot description format (URDF) model building

2.1. Four-wheel drive automated guided vehicle forward kinematic model

Forward kinematic analysis requires two coordinate systems: the map coordinate system and the vehicle coordinate system. The results of the forward kinematic measurements are the map coordinates (x, y, z) in the x-y-z axis and the Euler angles (roll, pitch, yaw) in the map coordinates for the three axes.

As shown in figure 1. AGV moves in the map coordinate system, α denotes moving yaw angle, $\Delta\alpha$ denotes instantaneous change in yaw angle, v denotes velocity, and d denotes displacement. AGV in the vehicle coordinate system, according to the X' of IMU as the forward direction, the right side is the Y' direction of IMU, θ denotes steering angle, in thex' direction displacement instantaneous change is $\Delta x'$, in the y' direction displacement instantaneous change is $\Delta y'$.

Instantaneous displacement of the whole car in the forward direction (X'-axis): the sum of the two rear wheel displacements is always equal to two times the displacement of the whole car.

$$\Delta x' = \frac{(\Delta d_1 + \Delta d_2)}{2}. \tag{1}$$

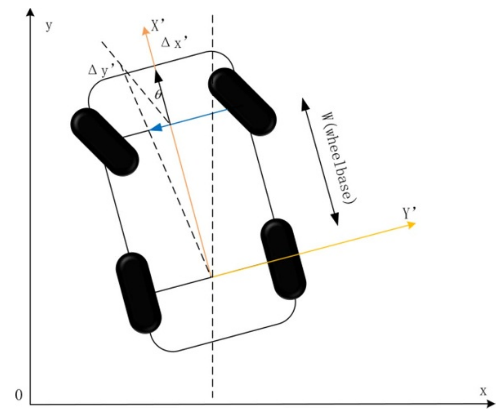

Figure 1. Four-wheel drive AGV forward kinematic model.

From the heading angle and the forward displacement of the whole vehicle, the cumulative displacement of the cart in the map coordinate system in the X-axis and Y-axis is calculated as follows:

$$x = x + \Delta x' \cos\alpha \tag{2}$$

$$y = y + \Delta y' \sin\alpha \tag{3}$$

(x, y) denotes the actual position of the cart in the map coordinate system, without considering the vertical direction so $z = 0$.

Instantaneous displacement of the AGV on the y'-axis of the vehicle coordinate system:

$$\Delta y' = \Delta x' \tan\theta. \tag{4}$$

Instantaneous angular velocity of AGV at $\Delta y'$ small enough:

$$\Delta a = \frac{\Delta y'}{W} = \frac{\Delta x' \tan\theta}{W}. \tag{5}$$

It can be further derived that the yaw angle in the map coordinate system:

$$a = a + \Delta\alpha. \tag{6}$$

To sum up the complete information of AGV posture (x, y, $z = 0$, roll $= 0$, pitch $= 0$, yaw).

2.2. Four-wheel drive automated guided vehicle URDF model

This article is based on the ROS system and Gazebo simulator to write a four-wheel drive AGV 3D model using URDF files. URDF is written in XML language to describe the basic structure of the desired model, joints, and sensors that ultimately make up the overall model.

The four-wheel drive AGV model includes the AGV body, differential wheels for travel, and steering. After completing the most basic model of the AGV, LIDAR sensors are added to sense the surrounding environment.

The basic parameters of the built AGV model are shown in table 1.

Table 1. AGV basic parameters.

Parameter	Numerical value
Complete vehicle weight (kg)	180
Wheel base (m)	1.1
Rolling radius (m)	0.22

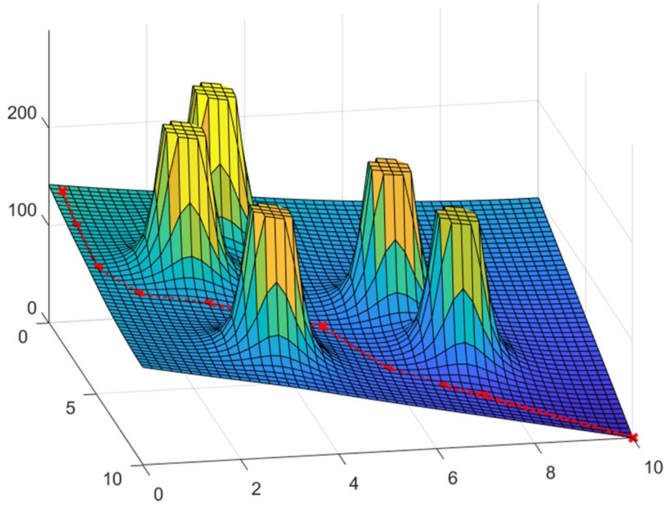

Figure 2. Principle of artificial potential field.

3. Algorithm

3.1. Traditional APF method

The basic idea of the APF algorithm is to use a method similar to the potential gravitational field, where obstacles are objects with mass exerting a repulsive force, and a target point is an object with mass exerting a gravitational force. As shown in figure 2, the AGV slides down from the highest to the lowest point to reach the target point.

The APF method simulates the environment around the AGV as a natural potential field, called a virtual potential field. The AGV gradually approaches the target point with the help of the gravitational field; meanwhile, the repulsive field acts on the AGV to make it bypass the obstacle.

The APF contains two parts, the gravitational field and the repulsive field, expressed as:

$$U(q) = U_{\text{attr}}(q) + U_{\text{rep}}(q). \tag{7}$$

The gravitational potential field positively correlates with the distance between the AGV and the target point. The farther the AGV is from the target point, the more gravity it is subjected to, while the closer the AGV is to the target point, the less gravity it is being forced to exert. The gravitational potential field function is shown in (8),

$$U_{\text{attr}}(q) = \begin{cases} \frac{1}{2}\lambda d(q, q_{\text{goal}})^2, d(q, q_{\text{goal}}) \leqslant d_{\text{thres}} \\ \lambda d(q, q_{\text{goal}}) d_{\text{thres}} - \frac{1}{2}\lambda d_{\text{thres}}^2, d(q, q_{\text{goal}}) > d_{\text{thres}} \end{cases} \tag{8}$$

$$d(q, q_{\text{goal}}) = \left\| q - q_{\text{goal}} \right\|_2 \tag{9}$$

The $U_{\text{attr}}(q)$ denotes the gravitational field of the target point; $d(q,q_{\text{goal}})$ denotes the Euclidean distance between the current position of the AGV and the target point.

When the current distance is less than the threshold, the farther the distance, the greater the gradient and the more gravitational force. When the distance exceeds the threshold, the greater the distance, the smaller the gradient and the less gravitational force.

The repulsion field:

$$U_{\text{req}}(q)=\begin{cases}\frac{1}{2}\mu\left(\frac{1}{D(q,q_{\text{obst}})}-\frac{1}{D_{\text{thres}}}\right)^2, D(q,q_{\text{obst}})\leqslant D_{\text{thres}}\\ 0, D(q,q_{\text{obst}})>D_{\text{thres}}\end{cases} \tag{10}$$

$$D(q,q_{obst})=\|q-q_{obst}\|_2. \tag{11}$$

The$U_{\text{req}}(q)$ denotes the repulsive field of the obstacle. μ denotes the repulsive field coefficient factor. $D(q,q_{\text{goal}})$ denotes the distance from the current position of the AGV to the target point.

3.2. Improved APF

The APF method has many drawbacks. When the combined force is zero, the AGV cannot determine the directions and cannot move forward or oscillate at the current position. When the gravitational and repulsive forces are in a line, it will be impossible to determine the direction and reach the destination. When the target point is reached, but the repulsive force is still present, the AGV cannot stop at the endpoint.

In this article, the repulsive field function is changed into two parts. The first part is pointed by the obstacle to the AGV, and the size is determined by the distance between the AGV and the obstacle and the distance between the AGV and the target point. This is shown by the red arrow F_{req1} in the figure below. The AGV points the second part to the target point, and the size is determined by the distance between the AGV and the obstacle and the distance between the AGV and the target point. This is shown by the red arrow F_{req2} in the figure below. These two repulsive forces together form the combined repulsive force F_{req}, as shown by the orange arrow in the figure below. This force is then combined with the gravitational force F_{attr}, shown by the blue arrow, to obtain the final force on the AGV.

The force analysis of the AGV in the repulsive field is shown in figure 3. The conventional APF method of the obstacle repulsive potential field adds a regulating factor D_g^n so that the repulsive force and the gravitational force are simultaneously reduced to zero only when the car reaches the target point, thus solving the problems of local optimization and target unreachability.

Improved repulsive field functions:

$$U_{\text{req}}(X)=\begin{cases}\frac{1}{2}\mu\left(\frac{1}{D(q,q_0)}-\frac{1}{D_{\text{thers}}}\right)^2 D_g^n & 0\leqslant D(q,q_0)\leqslant D_{\text{thres}}\\ 0 & D(q,q_0)\geqslant D_{\text{thres}}\end{cases} \tag{12}$$

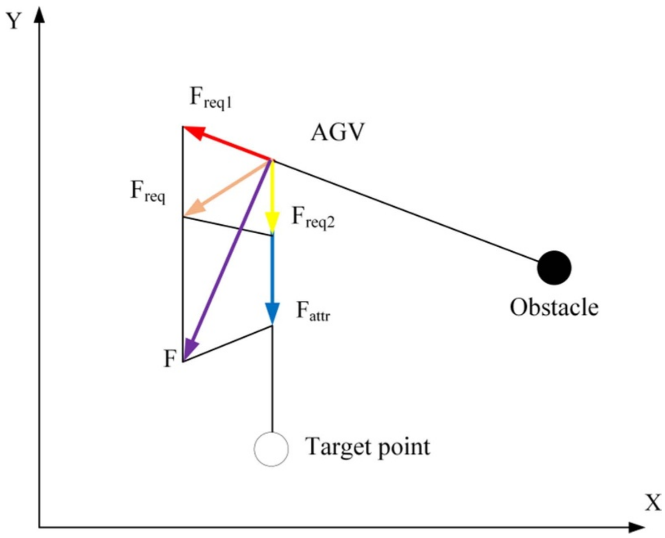

Figure 3. Force analysis of AGV in repulsive field.

$D(q,q_0)$ is the distance between the AGV and the obstacle, D_{thres} is the obstacle influence distance, and D_g^n is the distance between the AGV and the target point.

Compared with the traditional APF method, the distance between the AGV and the target point is added to the improved repulsive field function. This reduces the gravity force and the repulsion force on the AGV when driving to the target point to a certain degree simultaneously, i.e. the target point becomes the point with the smallest potential energy. When the AGV reaches the target point the repulsive force:

$$F_{\text{req}}(q)=\begin{cases}(F_{\text{req1}}+F_{\text{req2}})\nabla D(q,q_0) & 0\leqslant D(q,q_0)\leqslant D_{\text{thres}}\\ 0 & D(q,q_0)\geqslant D_{\text{thres}}\end{cases}. \tag{13}$$

With the above improvements, the AGV reaches the target point with a repulsive force of 0, as shown in (14).

$$\begin{cases}U_{\text{req1}}=\mu\left(\frac{1}{D(q,q_0)}-\frac{1}{D_{\text{thres}}}\right)\frac{D_g^n}{D^2(q,q_0)}\\ U_{\text{req2}}=\frac{n}{2}\mu\left(\frac{1}{D(q,q_0)}-\frac{1}{D_{\text{thres}}}\right)^2 D_g^{n-1}\end{cases} \tag{14}$$

$U_{\text{req1}}(q)$ and $U_{\text{req2}}(q)$ denote the repulsive field of the obstacle.

The traditional APF method is caught in a local optimum, as shown in figures 4 and 5, where the repulsive occasion force is equal in size to the gravitational force in the opposite direction. That is, $F_{\text{attr}}\leqslant F_{\text{req1}}+F_{\text{req2}}$.

Figures 6 and 7 show the improved APF method, which adds a regulating factor and a distance constraint to the repulsive field to solve the problem that the APF algorithm tends to fall into a local optimum.

3.3. Traditional A* algorithm

The traditional A* algorithm combines the advantages of the Dijksand breadth-first search algorithm. It is the most effective

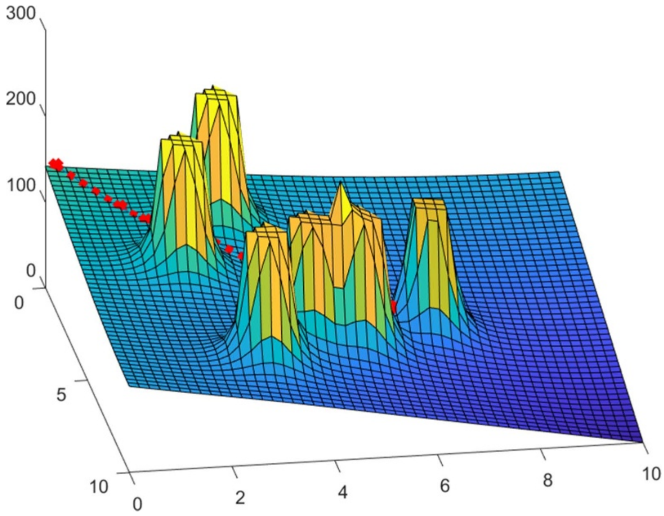

Figure 4. Lateral view of a traditional artificial potential field.

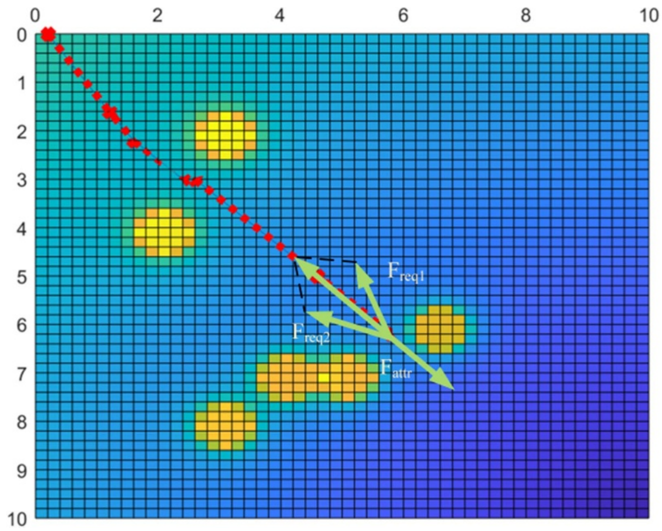

Figure 5. Overhead view of traditional artificial potential field.

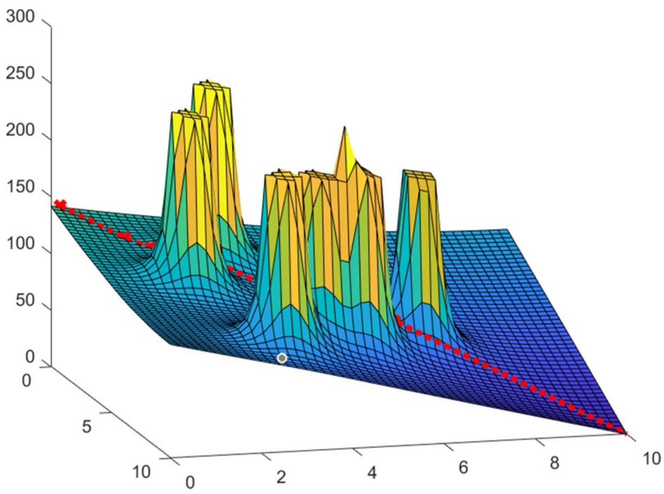

Figure 6. Lateral view of the improved artificial potential field.

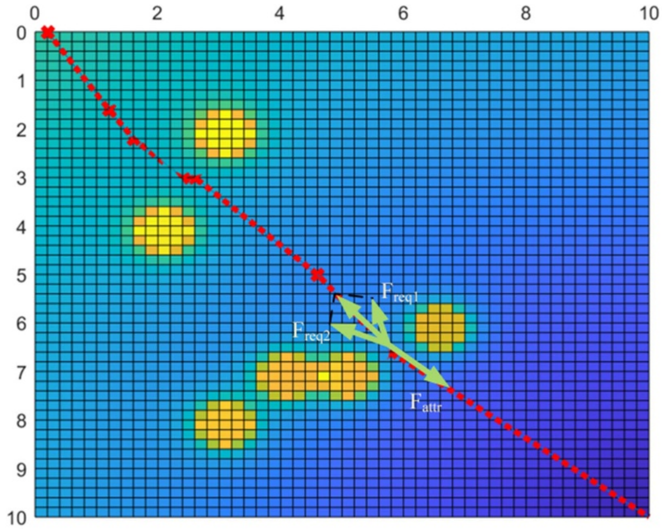

Figure 7. Overhead view of the improved artificial potential field.

heuristic search algorithm for solving the shortest trajectory in a static road network. The A* algorithm works by first searching in the state space and evaluating the cost function for each position searched. Then, it arrives at the best position as a child node and finally searches from this position until the endpoint.

The cost estimation function $f(n)$ from the initial state through n states to the target state is:

$$f(n) = g(n) + h(n) \tag{15}$$

$g(n)$ denotes the actual cost from the initial state to state n in the state space, $h(n)$ denotes the estimated cost from state n to the optimal trajectory of the target state. The closer the distance surrogate value in the algorithm is to the actual value, the faster the search will be.

3.4. Improved A* fusion algorithm by introducing APF

Since the A* algorithm does not consider redundant inflection points and unsmooth trajectories, the planned trajectories are smoothed, and redundant inflection points are removed by introducing the APF method. This paper introduces a gravity potential force field at the node where the target point is located in the grid graph to realize the goal of removing redundant nodes. A repulsive potential field is introduced at the node where the obstacle is located to make the angle of the AGV during steering operation more reasonable and to achieve the effect of the AGV driving trajectory smoothing. In this article, the gravitational and repulsive forces on each grid are superimposed, and then the combined force on that grid is calculated. The combined force is introduced into $g(n)$ of the A* algorithm so that the AGV can avoid being close to the obstacle while smoothing the trajectory, which is more in line with the actual driving needs. When the AGV is outside the influence of an obstacle in the grid map, it is subject only to the gravitational force of the target point and the repulsive force of the obstacle. When the AGV is within the influence of an obstacle in the grid map, it is subject to the gravitational force of the target point and the repulsive force exerted by the surrounding obstacles. So $g'(n)$ in the A* algorithm evaluation function is shown in (14):

$$g'(n) = g(n) + \sqrt{U_{\text{attr}}(q) + U_{\text{req1}}^2(q) + U_{\text{req2}}^2}. \tag{16}$$

The equation $U_{\text{attr}}(q)$ denotes the gravitational field of the target point.

The A* algorithm, after introducing of the APF, can effectively reduce the number of nodes in the Closed list during the search. The nodes covered by the repulsive field can avoid repeated searches by the A* algorithm, thus reducing the number of operations and improving the pathfinding speed. However, for the heuristic function of the A* algorithm, if the estimated value is smaller than the actual value, it will generate too many search nodes and reduce the computational efficiency, but the optimal trajectory can be obtained. If the estimated value is larger than the actual value, it will generate fewer search nodes, and although the efficiency is improved, the trajectory could be more optimal. Therefore, this article proposes to improve the A* algorithm heuristic function to balance this problem. At the beginning of the search phase, the estimator weights are increased to improve efficiency. During the search process, the estimated value weights continuously decrease as the target point is approached to obtain the best trajectory. Therefore, this paper introduces dynamic coefficients to the heuristic function of the A* algorithm. Compared with before the improvement, it will further reduce the number of search nodes in the closed list.

Add dynamic weight coefficients U to the cost function to improve the algorithm search efficiency trajectory cost function as in (17):

$$U = \frac{g(n)}{\sqrt{x_g - x_c + (y_g - y_c)^2}} \tag{17}$$

(x_c, y_c) denotes the coordinates of the initial node. (x_g, y_g) denotes the coordinates of the target node. As seen from (17), U is the ratio of the actual generation value $g(n)$ from the starting point to the current node and the distance from the starting point to the target node. Since the distance from the starting point to the target node is a constant value, the larger the value of $g(n)$ is, the larger the value of the dynamic parameter U is, and there is a positive relationship between the two. At the beginning of the trajectory exploration, the dynamic parameter U is also tiny because $g(n)$ is small, which makes the trajectory search unrestricted and can be explored diffusely. In the later stage of the trajectory search, the value of $g(n)$ is larger, which leads to the increase of the value of the dynamic parameter U. At this time, the range of the trajectory search is closer to the target node, which speeds up the efficiency of the trajectory search and the convergence ability of the algorithm is enhanced.

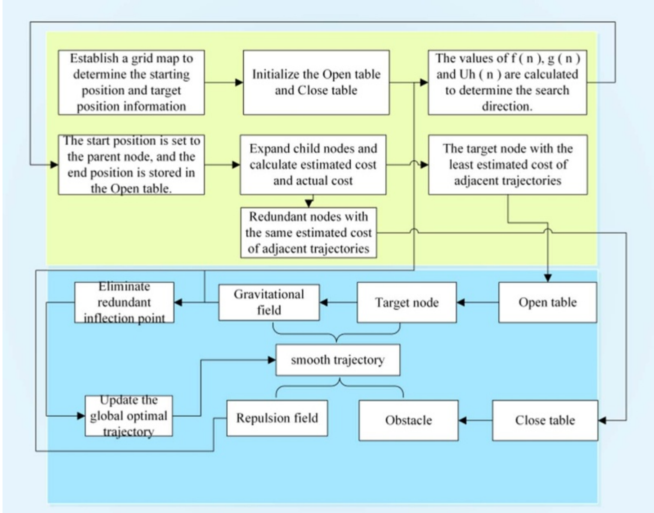

Figure 8. APF-A* algorithm procedure.

Adding dynamic weighting factors to the traditional A* algorithm:

$$f(n) = g(n) + Uh(n). \tag{18}$$

There are redundant search nodes, namely Openlist and Closelist nodes, which state that multiple nodes in the neighboring domain have the exact computationally intensive cost. The A* algorithm with an increased dynamic ratio increases the offset cost of preselected nodes to the target node. The redundant preselected nodes are removed.

The procedure of the APF-A* algorithm is shown in figure 8:

4. Simulation experiment and analysis

4.1. Algorithms comparison experiment

In this article, MATLAB is used as the algorithm simulation software. The simulation results of conventional A*, traditional APF algorithm, DWA-A*, and APF-A* algorithms are compared. The improved APF-A* is proved to have the advantages of faster search, shorter trajectory, and fewer redundant inflection points. As shown in figures 9–12:

The four algorithms for the same environment took the (40,13) coordinates as the starting point and the (28,14) coordinates as the ending point. The APF-A* optimized by (16) at (22,23) reduces three redundant inflection points compared with the conventional A* algorithm. The comparison yields the data shown in table 2. This article's proposed APF-A* algorithm optimizes 53% of the search cost, 57% of the required time, and 41% of the redundant inflection points compared to the traditional A* optimization. The APF-A* algorithm optimizes the search cost by 21% compared to the APF algorithm, optimizes the required time by 23%, and reduces the redundant inflection points by 32%. The APF-A* algorithm optimizes the search cost by 8.4% compared to the

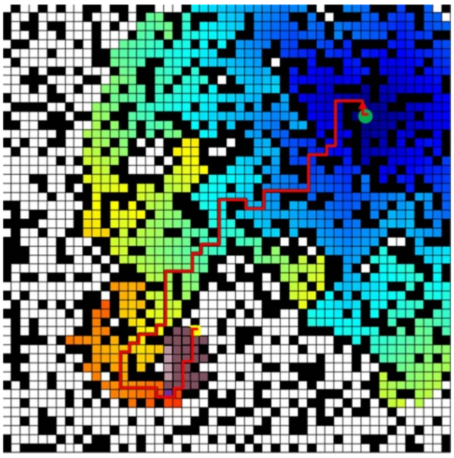

Figure 9. The traditional A* algorithm of environment A.

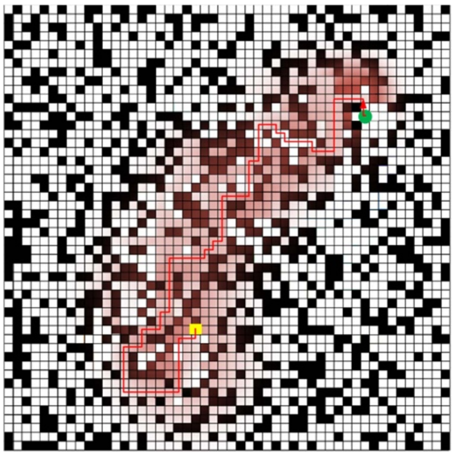

Figure 10. The traditional APF algorithm of environment A.

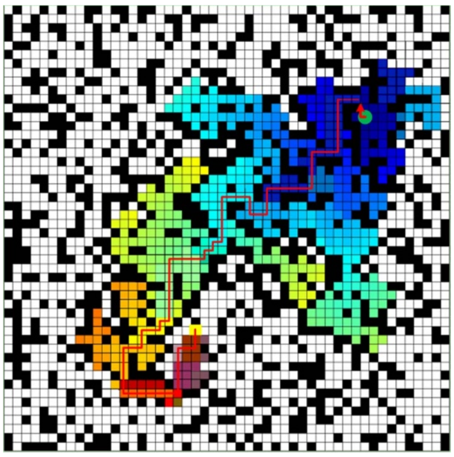

Figure 11. The DWA-A* algorithm of environment A.

DWA-A* algorithm, optimizes the required time by 72.4%, and reduces the redundant inflection points by 22.2%.

The four algorithms for the same environment took (26,46) coordinates as the starting point and (34,3) as the ending point. As shown in figures 13–16. The APF-A* optimized by (16) at (30,17) reduces one redundant inflection point compared to the traditional A* algorithm. The comparison yields the data shown in table 3. This article's proposed APF-A* algorithm optimizes 60% of the search cost, 68% of the required time, and 14% of the redundant inflection points compared to the traditional A*. The APF-A* algorithm optimizes the search cost by 48.7% compared to the APF algorithm, optimizes the required time by 56.4%, and reduces the redundant inflection points by 7.7%. The APF-A* algorithm optimizes the search cost by 6.3% compared to the DWA-A* algorithm, optimizes the required time by 39%, and reduces the redundant inflection points by 7.7%.

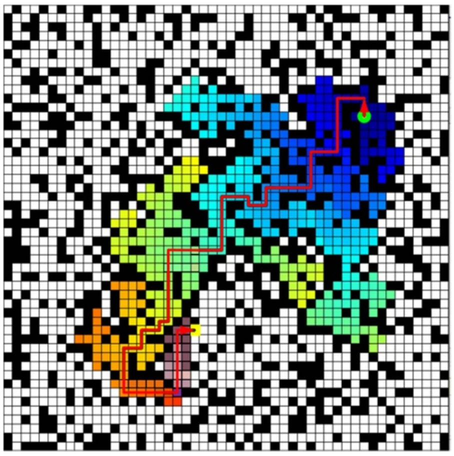

Figure 12. The APF-A* algorithm of environment A.

Table 2. Simulation data of environment A.

Algorithm	A*	APF	DWA-A*	APF-A*
Search Cost	956	563	486	445
Time (s)	4.246	2.362	2.544	1.820
Inflection Point	36	31	27	21

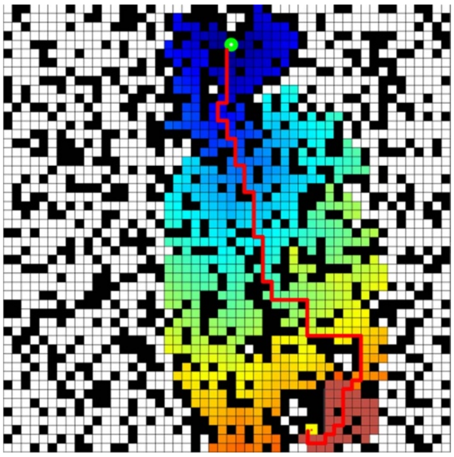

Figure 13. The traditional A* algorithm of environment B.

As shown in figures 17–19, comparing the simulation data in both environments shows that the difference between the improved DWA-A* and APF-A* is insignificant. However, the improved search cost of APF-A* is optimized by 60% compared to the traditional A*. The inflection points of different

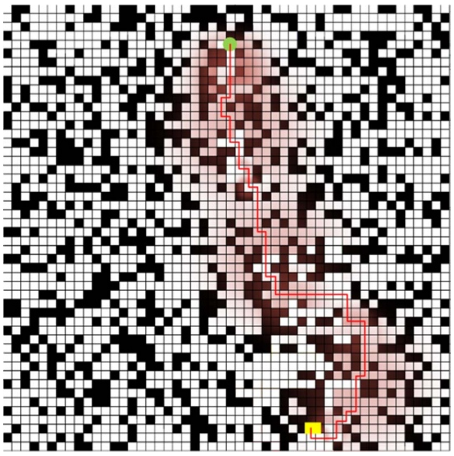

Figure 14. The APF algorithm of environment B.

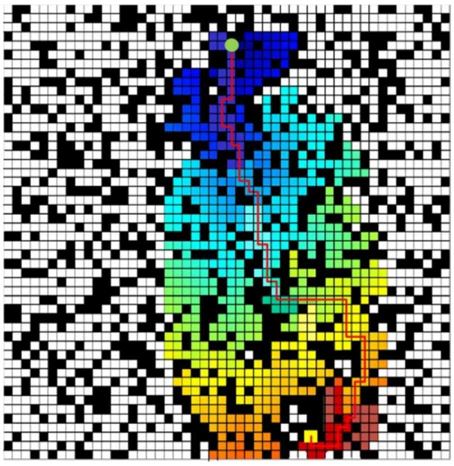

Figure 15. The DWA-A* algorithm of environment B.

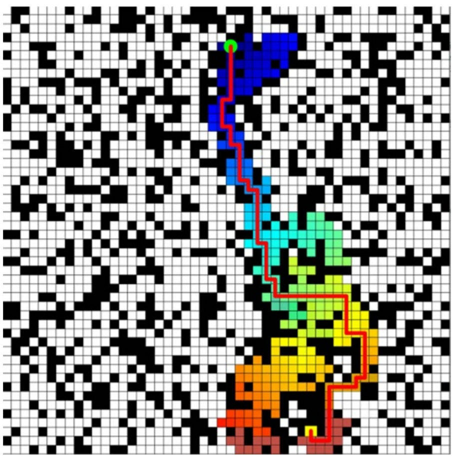

Figure 16. The APF-A* algorithm of environment B.

optimization critical nodes of the environment are different, and the number of optimization inflection points is also different. It can be concluded that APF-A* reduces the redundant inflection points by 14% $\sim$ 41% compared with traditional A*. The four algorithms, traditional A*, APF, DWA-A*, and APF-A*, reduce the search time in order. This article's APF-A* algorithm optimizes search time by up to 60% compared with the traditional A* algorithm.

Table 3. Simulation data of environment B.

Algorithm	A*	APF	DWA-A*	APF-A*
Search Cost	604	462	253	237
Time (s)	3.582	2.609	1.862	1.135
Inflection Point	28	26	26	24

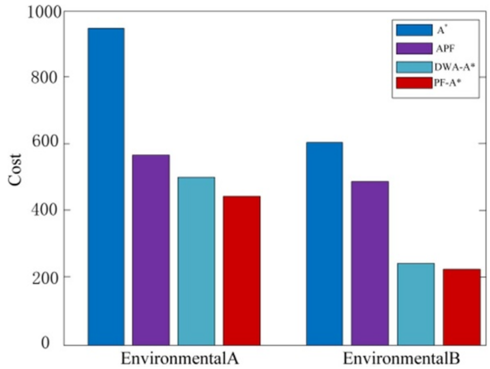

Figure 17. Comparison of simulation cost data in two environments.

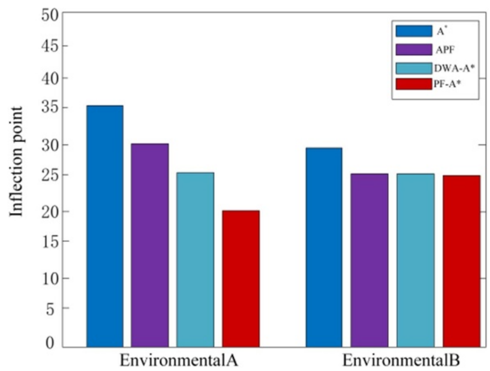

Figure 18. Comparison of simulation inflection point data in two environments.

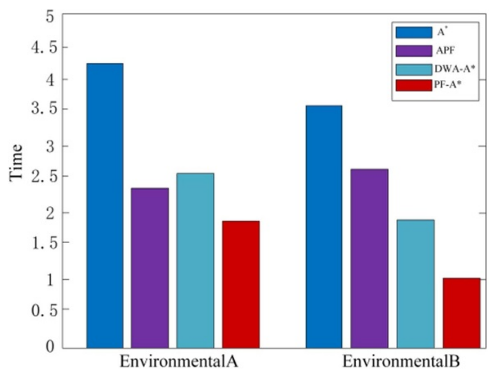

Figure 19. Comparison of simulation time data in two environments.

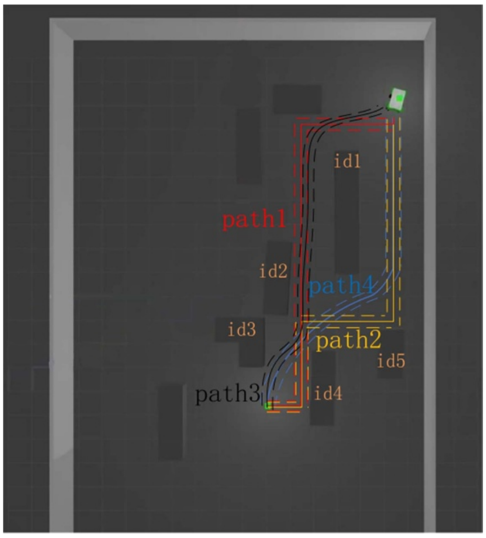

Figure 20. Simulation trajectory comparison.

Table 4. Obstacle parameters.

Parameter	Value
Id1	2.5×0.5 m
Id2	1.5×0.5 m
Id3	1.0×1.0 m
Id4	1.5×0.5 m
Id5	1.0×0.5 m
λ	50
d_{thres}	0.5 m

5. Automated guided vehicle trajectory simulation analysis

To verify the algorithm's optimization effect in this study, we perform simulation in ROS, that is, virtual environment navigation experiments. Simulation experiments based on an innovative factory environment. Based on the actual generation value of the algorithm $g(n)$ compare the search cost and time of the trajectory. The smoothness of traveling positional trajectories is compared based on the forward kinematic model of AGV in chapter 2. The optimization performance of AGV trajectory planning in terms of inflection points, search time, and search cost is evaluated. The intelligent factory environment with simulated multiple static obstacles is shown in figure 20. The obstacle parameters are shown in table 4.

Meanwhile, according to the actual environment, build the corresponding simulation environment in Gazebo and import the URDF model into the map as in figure 20. Path 1 in orange denotes the conventional A* planning trajectory. Path 2 in red denotes the APF planning trajectory. Path, 3 in black, denotes the DWA-A* planning trajectory. Path 4 in blue denotes the improved A* planning trajectory when introducing the APF. The comparison yielded the data shown in table 5.

Figure 20 shows a schematic diagram of the experimental plane, the experimental environment as a 10×9 m room, and the AGV speed as 4 m s^{-1}.

Table 5. Simulation data comparison.

Algorithm	A*	APF	DWA-A*	APF-A*
Search cost	700.772	680.440	674.582	658.689
Time (s)	8.945	6.382	5.463	3.578
Inflection point	3	3	1	0

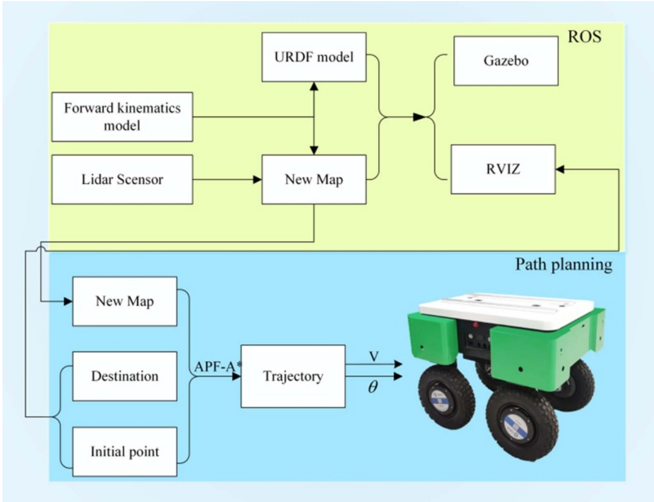

Figure 21. Experiment flow.

As shown in figure 20, the traditional A* algorithm trajectory is not smooth, has many inflection points, and has yet to reserve a safe distance from the obstacles. This article introduces the APF-A* algorithm, which introduces an improved APF method based on increased dynamic weights to reduce redundant inflection points, smooth trajectories, and collision hazards. Table 5 shows that the APF-A* algorithm studied in this article optimizes 6% of the search cost, 100% of the redundant inflection points, and 60% of the time compared to the traditional A* algorithm. Compared with the APF algorithm, the search cost is optimized by 3%, the redundant inflection point is optimized by 100%, and the time is optimized by 43.93%.

5.1. Experimental platform construction

The overall flow of the experiment in this paper consists of two main parts, as shown in figure 21. In the ROS layer, the URDF model is first built from the forward kinematic model, and the LIDAR scans the new map in the experiment. It is then configured and displayed in Gazebo and RVIZ. In the trajectory planning layer by the above map, start point and end point information through the APF-A* algorithm proposed in this article for trajectory planning.

In order to verify the practicality of the APF-A* algorithm, this article uses the self-developed 4WIS/4WID AGV for the actual vehicle test, and the real vehicle experimental platform is shown in figure 22. GPS/INS/IMU sensors obtain the vehicle's 3D position, attitude, velocity, and acceleration. The AGV is equipped with an industrial control

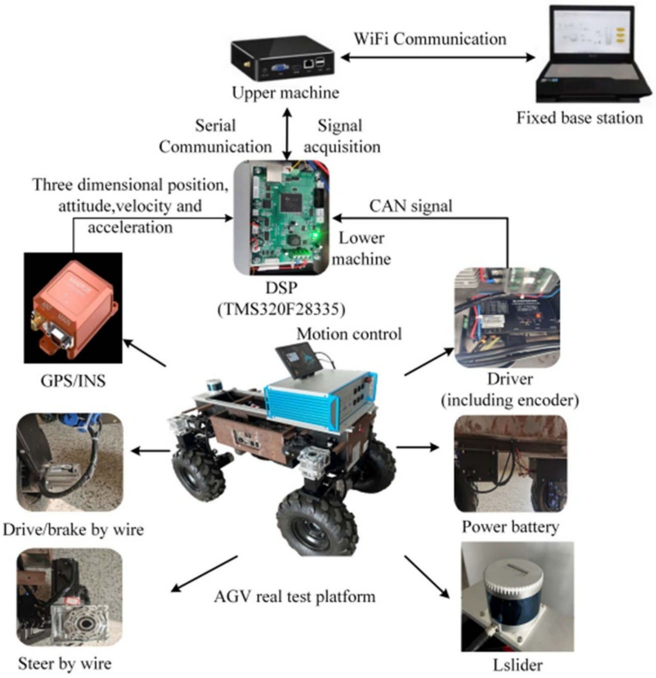

Figure 22. The real vehicle experimental platform.

Figure 23. Top view of experimental environment.

computer based on the ROS system to realize the upper-level intelligent control algorithm and connects to the base station through WIFI to realize the data collection and processing.

This article selects an indoor area as the experimental site for AGV trajectory planning, and chairs are used to simulate obstacles. As shown in figure 23, the scenario simulates a comprehensive utilization scenario of AGV in intelligent storage.

As shown in figure 24, the APF-A* algorithm is used for trajectory planning in the constructed map environment. The

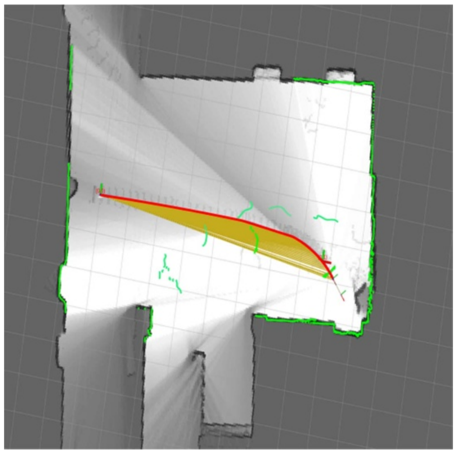

Figure 24. APF-A* algorithm experimental results.

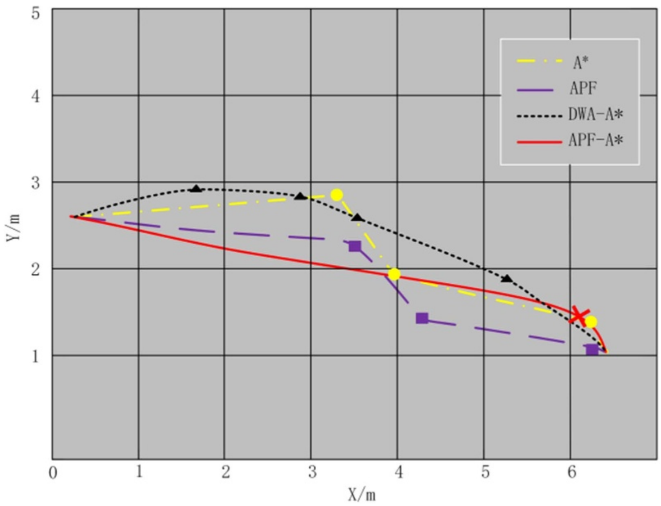

Figure 25. Comparison of four algorithmic paths.

green curve denotes the obstacles scanned by LIDAR, the yellow curve denotes the algorithm search cost curve, and the red curve denotes the path traveled by the AGV.

In the same environment, the traditional A*, APF, and DWA-A* algorithms are used for trajectory planning, respectively.

The path curves for the four algorithms are shown in figure 25. The yellow curve indicates the traditional A* algorithm. The planned path contains three inflection points and is labeled with '●'. The purple curve represents the traditional APF algorithm where the planned path contains three inflection points labeled with '■'. The black curve represents the DWA-A* algorithm. The planned path contains four inflection points and is labeled with '▲'. The red curve represents the APF-A* algorithm. The planned path contains one inflection point marked with an '✖;'. The APF-A* algorithm has the shortest path length, the most minor search cost, and the shortest planning time.

Table 6. Comparison of experimental data.

Algorithm	A^*	APF	DWA-A^*	APF-A^*
Search cost	600	546	532	522
Time (s)	2.630	1.986	1.583	1.052
Inflection point	3	3	4	1

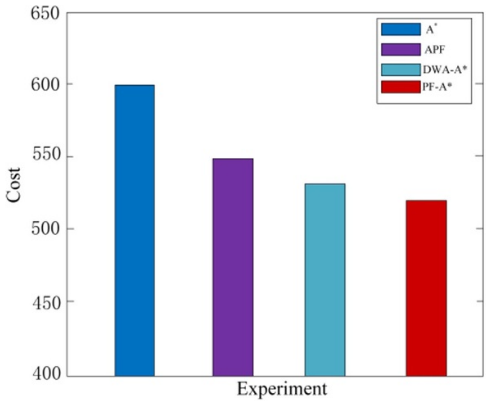

Figure 26. Comparison of costs in the experiment.

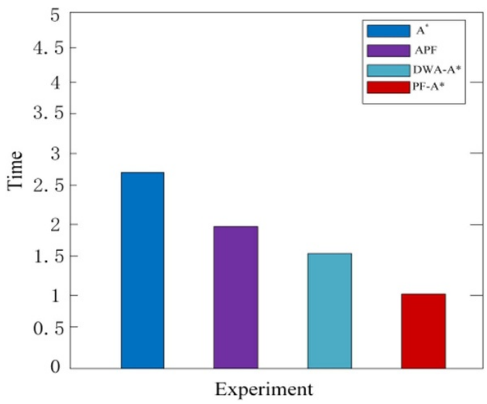

Figure 27. Comparison of time in the experiment.

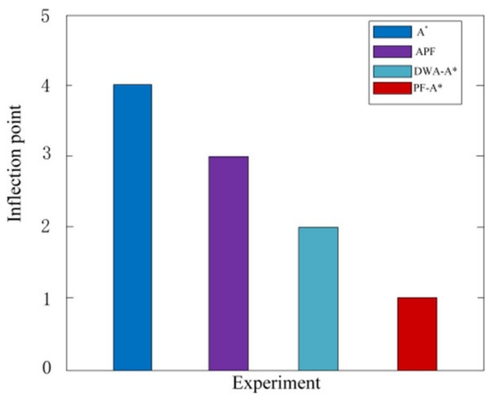

Figure 28. Comparison of inflection points in the experiment.

Table 6 shows the results of the experimental parameters of the three algorithms shown in figures 26–28.

As shown in figures 26–28, The APF-A^* algorithm proposed in this article optimizes the search cost by 13%, the redundant inflection points by 75%, and the time duration by 60% compared with the traditional A^* algorithm. The APF-A^* algorithm optimizes 50% of the redundant inflection points and 33.6% of the time duration compared to the DWA-A^* algorithm. The traditional A^* algorithm has high search costs, multiple inflection points, and abrupt changes in trajectory. The APF-A^* algorithm proposed in this article introduces an improved APF method based on increasing the dynamic weights to smooth the trajectory, reduce the redundant inflection points, effectively shorten the trajectory length, and increase the planning efficiency.

6. Conclusion

This article addresses the A^* algorithm and the APF method. Among them, the A^* algorithm has problems such as great computational effort and unsmooth trajectory curves. The APF method is prone to the problem of local minima in trajectory planning, which makes the trajectory planning incomplete. This article improves a method of dynamic weight coefficients to further optimize the efficiency of trajectory planning and the trajectory curve. After improving the A^* algorithm, it integrates the A^* algorithm and the APF algorithm, which can effectively reduce the A^* algorithm's search nodes.

The APF-A^* algorithm transforms the weights of the A^* algorithm into dynamic weights. It combines A^* with the APF method so that the two algorithms complement each other. Among them, the A^* algorithm realizes real obstacle avoidance, and the APF method accelerates the construction of trajectory curves in the blank area, eliminates redundant inflection points, and smooths the curves of critical nodes. MATLAB simulation experimental results show that the trajectory curve searched by the APF-A^* algorithm is more suitable for the actual motion of AGV than the traditional A^* algorithm, which smooths the trajectory and reduces the redundant inflection points of the trajectory to a certain extent. The APF-A^* algorithm reduces the trajectory search time by 60%, the search cost by 6%–13%, and the trajectory redundant inflection points by 75%–100% compared with the traditional A^* algorithm, indicating that the trajectory planned by the APF-A^* algorithm performs better. Compared to the DWA-A^* algorithm, the APF-A^* algorithm reduces the trajectory search time by 33.6%, the search cost by 6%—13%–8.4%, and the redundant inflection points of the trajectory by 75%—100%–50%, indicating that the trajectories planned by the APF-A^* algorithm are more efficient.

Overall, the APF-A^* algorithm proposed in this paper is highly efficient in multi-static obstacle environments and shows strong adaptability. It solves the limitations of mainstream algorithms and can be widely used in practical scenarios such as smart factories and fruit-picking gardens. However, AGV also needs local path-planning algorithms to avoid sudden unknown obstacles. In the future, we should research path planning for AGVs in dynamic environments. In terms of AGV localization and dynamic obstacle avoidance, a

UWB base station can be realized by combining differential methods. In the process of path planning, the real-time progress display of a PC can also be considered, which can help users understand the working status of AGV more clearly.

Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

Declarations

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